Lecture 18: Parametrizations of Plane Curves

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

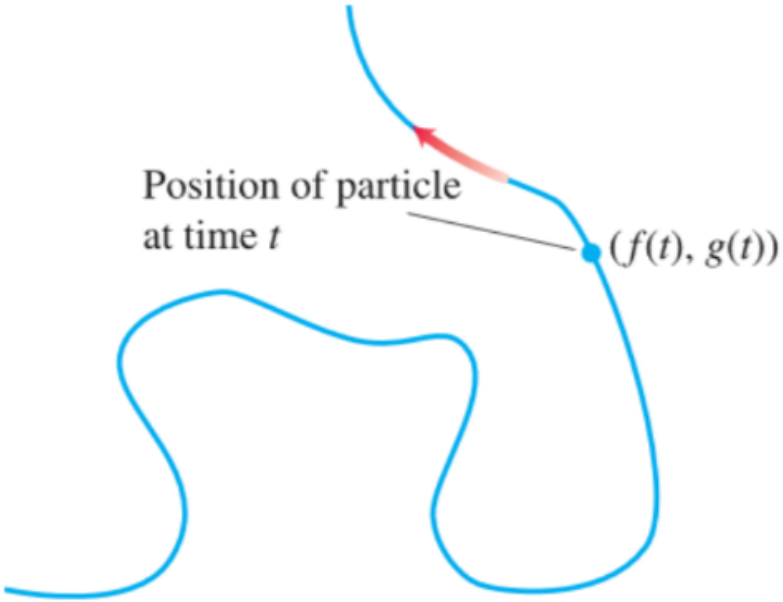
At the end of this lecture, you should be able to:

- ☐ Understand the concept of a curve defined parametrically.
- ☐ Convert between parametric and Cartesian equations when possible.
- ☐ Sketch curves given by parametric equations and describe their orientation.
- ☐ Analyze the behavior of a parametric curve over a given interval of the parameter.
- ☐ Identify and graph parametric representations of common and piecewise curves.

Parametric Equations

We sometimes define relation between x and y in terms of a third party called a parameter.

Instead of describing a curve as the graph of a function, we think of it as a path of a moving particle whose position is changing over time.



Parametric Equations

If x and y are given as functions

$$x = f(t), \qquad y = g(t)$$

over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a parametric curve. The equations are parametric equations for the curve.

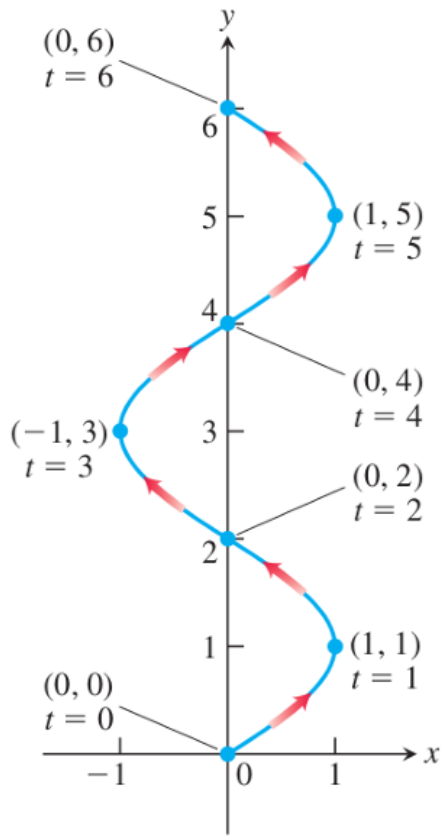
Parametric Equations

Example: Sketch the curve defined by the parametric equations

$$x = \sin\frac{\pi t}{2}, \quad y = t, \quad 0 \le t \le 6.$$

TABLE 10.1 Values of $x = \sin \pi t/2$ and $y = t$ for selected values of t.

t	x	y
0	0	0
1	1	1
2	0	2
3	−1	3
4	0	4
5	1	5
6	0	6



Parametric Equations

Example: Sketch the curve defined by the parametric equations

$$x = t^2, \quad y = t + 1, \quad -\infty < t < \infty.$$

TABLE 10.2 Values of $x = t^2$ and $y = t + 1$ for selected values of t.

t	x	y
-3	9	-2
-2	4	-1
-1	1	0
0	0	1
1	1	2
2	4	3
3	9	4

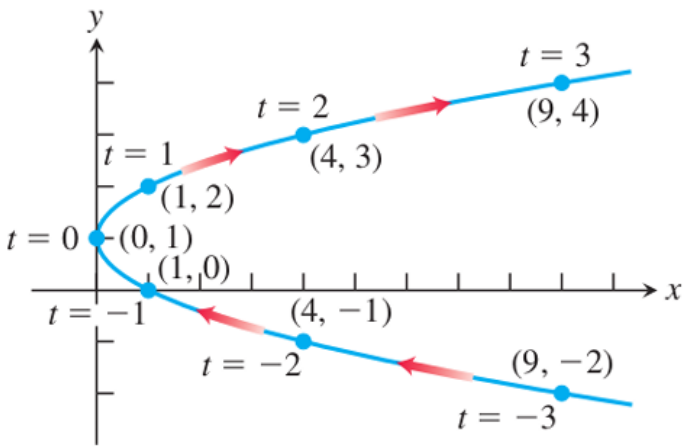


FIGURE 10.3 The curve given by the parametric equations $x = t^2$ and $y = t + 1$ (Example 2).

Parametric Equations

Example: Graph the parametric curves

(a) $x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi.$

(b) $x = a\cos t, \quad y = a\sin t, \quad 0 \leq t \leq 2\pi.$

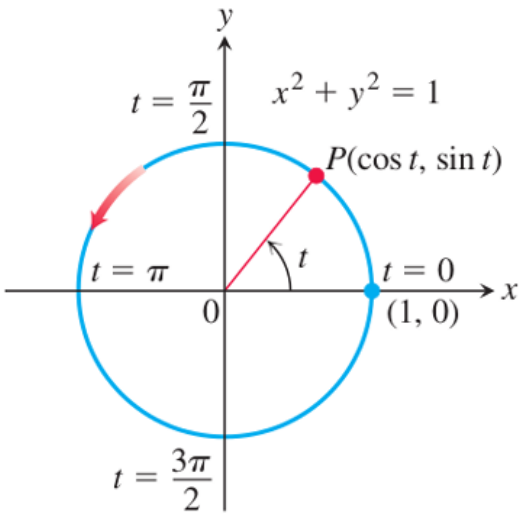


FIGURE 10.4 The equations $x = \cos t$ and $y = \sin t$ describe motion on the circle $x^2 + y^2 = 1$. The arrow shows the direction of increasing t (Example 3).

Parametric Equations

Example: The position $P(x, y)$ of a particle moving in the xy-plane is given by the equations and parameter interval

$$x = \sqrt{t}, \quad y = t, \quad t \geq 0.$$

Identify the path traced by the particle and describe the motion.

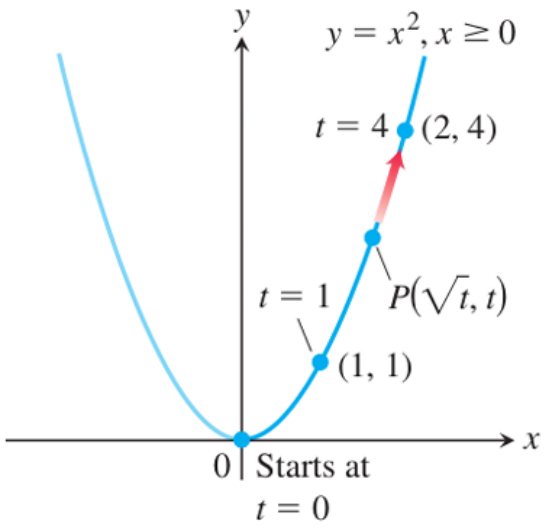


FIGURE 10.5 The equations $x = \sqrt{t}$ and $y = t$ and the interval $t \geq 0$ describe the path of a particle that traces the right-hand half of the parabola $y = x^2$ (Example 4).

Parametric Equations

Example: Find a parametrization for the line through the point (a, b) having slope m.

Parametric Equations

Example: Sketch and identify the path traced by the point $P(x, y)$ if

$$x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}, \quad t > 0.$$

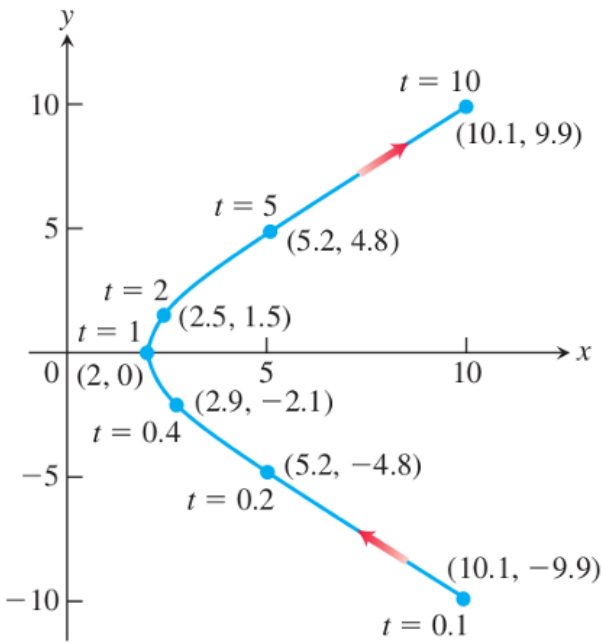


FIGURE 10.7 The curve for $x = t + (1/t)$, $y = t - (1/t)$, $t > 0$ in Example 7. (The part shown is for $0.1 \le t \le 10$.)

Parametric Equations

Any function can be rewritten parametrically in many different ways

Example: $y = x^3 + 7$.

Parametric Equations

Cycloids

A wheel of radius a rolls along a horizontal straight line. Find the parametric equations for the path traced by a point P on the wheel's circumference. The path is called a cycloids.

